Band-1 Farrell–Jones: Group-Class Lit Probe (Collapse 2.34)

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2026-09-11

Band-1 Farrell–Jones — One Group Class (Lit Lock)

Honesty stamp. This note does **not** claim the Farrell–Jones conjecture for all groups. It does **not** claim a Clay-millennial solution or a Theory of Everything. **GATE-04** is locked out of scope. Lit lock only — **no** assembly-map / controlled-algebra computation on box.

> **Locked PROVED (Band-1 class): hyperbolic groups** (Gromov) — K- and L-theoretic Farrell–Jones with coefficients (**Bartels–Lück–Reich** line).
> **Locked OPEN:** $\mathrm{Out}(F_n)$ for finite rank $n \geq 3$ — outer automorphism groups of free groups — **OPEN** (still called a major open problem in **Math. Ann. 2026**).
> **Explicit non-claim:** full FJ for **all** groups — **not** asserted.

Abstract

The **Farrell–Jones conjecture** (FJ) predicts that the algebraic K- and L-theory of group rings (and more generally of additive categories with G-action) is determined by an assembly map from the classifying space for virtually cyclic subgroups. A large **class FJ** of groups is known to satisfy the (full / fibered) conjecture — including **hyperbolic** groups, finite-dimensional **CAT(0)** groups, **virtually solvable** groups, **mapping class groups**, many **lattices** / S-arithmetic groups, and (as of 2026) $\mathrm{Aut}(G)$ / $\mathrm{Out}(G)$ for one-ended hyperbolic G. The conjecture remains **open** for several named classes; among them, $\mathrm{Out}(F_n)$ is still explicitly flagged as a major open problem in 2026 literature.

This Band-1 probe locks **one proved group class** (hyperbolic) and **one open group class** ($\mathrm{Out}(F_n)$). It does **not** elevate membership of hyperbolic groups (or any finite list) to an all-groups claim.

Problem (collapse Band-2 item 2.34 → Band-1 lit lock)

1. Ledger Farrell–Jones status; name proved classes and open remainders (2024–2026 check).
2. Lock **one** literature-proved group class as the Band-1 claim surface.
3. Name **one** remaining open group class with a recent citation.
4. Keep Clay / ToE / GATE-04 / "FJ for all groups" off the claim surface.
5. Lit lock enough — no box assembly computation.

Canonical statement (sketch)

For a group G and an additive category $\mathcal{A}$ with right G-action, the **K-theoretic Farrell–Jones assembly map** (virtually cyclic family $\mathcal{VC}$) is

$$H_n^G\big(E_{\mathcal{VC}}G;\, \mathbf{K}_{\mathcal{A}}\big) \;\longrightarrow\; K_n\big(\mathcal{A} *_G G\big).$$

FJ asserts this is an isomorphism for all $n \in \mathbb{Z}$ (and likewise for L-theory; fibered / "full" variants strengthen coefficients and inheritance). Equivalent classical forms recover Whitehead groups, K_0 of group rings, and surgery obstructions from equivariant homology.

Literature ledger (web check 2024–2026; not re-proved on box)

Year	Result	Authors / venue	Status
1993+	Conjecture formulation (assembly)	Farrell–Jones	classical statement
2008	**K-theoretic FJ for hyperbolic groups** (coefficients)	**Bartels–Lück–Reich**, *Invent. Math.*	**PROVED — Band-1 class**
2008–12	L-theory / full FJ for hyperbolic; CAT(0)	BLR; Wegner; Bartels–Lück	**proved**
2015	Virtually solvable groups	Wegner	**proved**
2014–16	$\mathrm{GL}_n(\mathbb{Z})$, lattices, S-arithmetic	BLRR; KLR; Rüping	**proved** (large linear / arithmetic)
2019	Mapping class groups	Bartels–Bestvina	**proved**
2023	Suspensions of v.t.f. hyperbolic groups	Bestvina–Fujiwara–Wigglesworth	**proved**
2026	Aut of one-ended RH-to-v.polycyclic; suspensions of many RH	*Math. Ann.* (open access, 16 Jun 2026)	**proved** (new Aut class); **explicitly does not settle** $\mathrm{Out}(F_N)$
2026	FJ for localising invariants / richer coefficients	*Adv. Math.* 2026	coefficients/functors — **not** a new all-groups proof
—	$\mathrm{Out}(F_n)$, $n \geq 3$	—	**OPEN**
—	Thompson groups; general elementary amenable beyond v.solvable; arbitrary linear; infinite products	surveys (e.g. Reich–Varisco style lists)	**OPEN** (partial injectivity only in places)

Survey pointers. Lück, *Survey on the Farrell–Jones conjecture*, Bull. AMS; Lück, *Assembly Maps* notes (class **FJ** inheritance list: hyperbolic, finite-dim CAT(0), virtually solvable, lattices, π_1 of $\leq$ 3-manifolds, $\mathrm{GL}_n(\mathbb{Q})$, S-arithmetic, mapping class groups; closed under subgroups, finite products, directed colimits, etc.).

What is proved vs open (statement lock)

Literature-proved — Band-1 locked class

1. **Hyperbolic groups** — Bartels–Lück–Reich: K-theoretic Farrell–Jones with coefficients; L-theoretic and full/fibered forms in the same program. This is the collapse-2.34 "new group class" archetype that opened the modern controlled-algebra era. **Locked PROVED class for this probe.**
2. **Context (also proved; not the single lock):** finite-dimensional CAT(0); virtually solvable; mapping class groups; many lattices / S-arithmetic / $\mathrm{GL}_n(\mathbb{Z})$; free-by-cyclic and hyperbolic-by-cyclic; Aut/Out of one-ended hyperbolic groups (2026).

Locked OPEN — named remaining class

1. $\mathrm{Out}(F_n)$ **for** $n \geq 3$ — **OPEN**. The June 2026 *Mathematische Annalen* paper on automorphisms of relatively hyperbolic groups states explicitly that whether $\mathrm{Out}(F_N)$ satisfies $\mathbf{FJC}_X$ is a **major open problem**, and that their theorems do **not** solve it (F_n is infinitely ended; the new Aut results cover one-ended hyperbolic / RH-to-v.polycyclic cases).
2. **Also open (not the single named lock):** Thompson groups; elementary amenable groups beyond the virtually solvable case; arbitrary linear groups; infinite products of FJ groups.

Explicit non-claims

- Farrell–Jones **not** claimed for **all** groups.
- $\mathrm{Out}(F_n)$ **not** claimed proved.
- No Clay; no ToE; **GATE-04 LOCKED**.
- No box assembly numerics presented as evidence.

Why Band-1 vs full conjecture

- **Band-1 (this probe):** honest lit lock — one proved class (hyperbolic) + one open class ($\mathrm{Out}(F_n)$), with 2024–2026 citation hygiene.
- **Programmatic next increments:** further published inheritance / Aut extensions (as in 2026 Math. Ann.), still without an all-groups claim.
- **Band 3–4:** FJ for all groups (or a uniform geometric criterion covering $\mathrm{Out}(F_n)$, Thompson, etc.).

Box note

No controlled N-ε algebra, no assembly-map spectral sequence, no Flow-space construction on box. Dash rule: **lit lock is enough**.

Deliverables

- Lab: `experiments/band1/farrell_jones/` (`literature_ledger.json`, `results.json`, `SUMMARY.txt`, `METHOD.md`, `README.md`, local `next_wave.md`).
- Paper/PDF: `papers/2026-09-11_Band1_Farrell_Jones_group_class_probe.{md,pdf}` → `DOWNLOAD_PDFs` + Mac `Downloads/DOWNLOAD_PDFs_2026-09-11/`.
- House `next_wave.md` updated.

References (primary)

1. F.T. Farrell, L.E. Jones, *Isomorphism conjectures in algebraic K-theory*, J. Amer. Math. Soc. **6** (1993).
2. A. Bartels, W. Lück, H. Reich, *The K-theoretic Farrell–Jones Conjecture for hyperbolic groups*, Invent. Math. **172** (2008), 29–70.
3. W. Lück, *Survey on the Farrell–Jones conjecture*, Bull. Amer. Math. Soc. (survey).
4. W. Lück, *Assembly Maps* (notes; class **FJ** list and inheritance).
5. C. Wegner, *The Farrell–Jones conjecture for virtually solvable groups*, J. Topol. **8** (2015).
6. A. Bartels, M. Bestvina, *The Farrell–Jones Conjecture for mapping class groups*, Invent. Math. **215** (2019).
7. M. Bestvina, K. Fujiwara, D. Wigglesworth, *Farrell–Jones conjecture for free-by-cyclic groups / suspensions of hyperbolic groups* (BFW line, 2019–2023).
8. *Automorphisms of relatively hyperbolic groups and the Farrell–Jones conjecture*, Math. Ann. (2026), open access 16 Jun 2026 — proves Aut/Out for one-ended RH-to-v.polycyclic; **explicitly leaves** $\mathrm{Out}(F_N)$ **open**.
9. *On the Farrell–Jones conjecture for localising invariants*, Adv. Math. (2026) — coefficient/functor extensions.